24 Learning by Diffusion: Using Diffusion Experiments and Social Network Analysis to Understand the Dynamics of Cultural Evolution

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Abstract

Culture arises from the interaction of many individuals sharing knowledge and collaborating over time, and, because of this, culture must be studied using different methods to those that are commonly employed in many areas of psychology. The majority of our understanding of the social learning underpinning culture is generalised from 'dyadic' experiments, in which a single participant observes a single model, and as a result leaves questions about the relationship between the individual- and group-level unaddressed. Such questions include how different forms of cultural information spread across the population and how individuals work together to produce cultural products. Diffusion experiments present a method for such dynamics to be examined. This chapter reviews the varieties of diffusion methods available and the strengths and weaknesses each type of diffusion design provides in answering questions about cultural evolution. It also reviews recent innovations in studying the spread of culture via social relations using social network analyses. We argue that social network analyses could be especially useful for examining a dynamic which has hitherto not been widely considered in studying cultural evolution; that is, the feedback relationship between social structure (relations) and social learning, with copying behaviour fulfilling both a role of information exchange and a role of affiliation.

Humans exhibit remarkable biological success as a species; we have undergone an extraordinary demographic shift, and inhabit every corner of the globe (Hill, Barton, & Hurtado, 2009). Yet this success cannot be solely attributed to inborn cognitive abilities; highly intelligent and well-resourced explorers have failed to survive in new terrains, where indigenous populations thrive (Henrich & McElreath, 2003). Human's biological success must also be understood with reference to culture, which is fundamental in constructing the cognition of individual humans. Culture arises out of the interaction of

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individuals and is a group-level phenomenon, which means that it is particularly difficult to investigate using traditional methods of experimental psychology. Yet as culture plays such an important part in our cognition, as well as our ultimate success as a species, it is essential to develop techniques that allow the interplay between individuals and groups to be explored to address the central questions of how culture is acquired, how it is transmitted and how it changes or evolves. By allowing an experimenter to control and measure both group and transmission processes, diffusion experiments offer a powerful experimental method for addressing these questions. This chapter reviews the diffusion method, and points to a necessary area of extension: namely, the feedback relationship between social structure and culture. Additional methods which offer promise in addressing the complexity in these dynamics are highlighted.

Culture and Human Cognition

Human life is predicated on the successful acquisition of cultural knowledge (Boyd, Richerson, & Henrich, 2011; Hill, Barton, & Hurtado, 2009; Richerson & Boyd, 2005; Sterelny, 2003). While non-human animals (henceforth, 'animals') possess abilities for culture and traditions, even in our most closely related relative, the chimpanzee (Whiten, 2011), the diversity and elaboration of cultural products pales in comparison to that found in humans. Further, human culture has the distinctive attribute of being cumulative: with cultural products and knowledge accruing advantageous modifications over generations, allowing increasing adaptive-fit with the environment over time (Dean, Vale, Laland, Flynn & Kendal, 2014; Tomasello, 1999; Tomasello, Kruger, & Ratner, 1993). Because of the cumulative properties of human culture, it has been argued that – for humans – culture constitutes a second inheritance system working in parallel with genetic inheritance and undergoing its own evolutionary change (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981; Flynn, Laland, Kendal, & Kendal, 2013; McElreath & Henrich, 2007). That is, along with the genes passed down by parents to offspring, humans also pass down a store of knowledge and artefacts to their children, who will then improve these cultural products, transmitting the modified versions to their own children, and so on.

Fundamental to culture is the transmission of information between individuals. The cognitive abilities and proclivities for social learning, known as 'social learning mechanisms', are key to understanding culture (Flynn & Whiten, 2013; Hoppitt & Laland, 2013; Zentall, 2006, 2012). As children are 'cultural magnets', the development of these special capacities for social learning are especially important (Flynn, 2008). The development of these cognitive capacities early in childhood allow humans to learn the distinctive cultural

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knowledge of their group; knowledge which has been accrued over generations to allow success in their particular environment (Tomasello, 1999; Tomasello et al., 1993). Cumulative culture in particular requires high fidelity transmission of information across generations (Lewis & Laland, 2012), and as children are the link between generations, studying children's capacities for high fidelity transmission sheds light on how human culture emerged out of the cultural capacities possessed by other apes (Horner, Whiten, Flynn, & de Waal, 2006; van Schiak & Burkart, 2011; Tomasello, 1999; Tomasello et al., 1993, Whiten, 2011).

Indeed, many studies now affirm a substantial difference in the abilities and motivation of young children and other closely related species in the domain of social learning. For instance, Herrmann, Call, Hernàndez-Lloreda, Hare, Hare and Tomasello (2007) compared the performance of children (two-year-olds) to that of chimpanzees and orangutans on a range of behavioural tests of intelligence. On tests which were within the physical domain, such as reasoning about space and causality, all groups performed equally. However, in tests within the social domain, such as following gaze, children outperformed non-human primates. Dean, Kendal, Schapiro, Theirry, and Laland (2012) compared children, chimpanzees and capuchin monkeys on the ability to extract rewards from a complex apparatus which required problem solving to advance through three stages, where each stage provided an increasingly desirable reward. It was found that children more often reached the later stages of this apparatus than the chimpanzees or the capuchin monkeys, and this was shown to be attributed to their greater use of social learning.

Early theorists in developmental psychology highlighted the situated nature of the development of human cognition within a cultural history, and the manner in which cognition is substantially constructed by social learning to acquire this cultural knowledge (Bandura, 1977; Vygotsky, 1978). Contemporary research confirms culture can have pervasive effects on cognition, with differences existing between groups, not only regarding beliefs and attitudes, but also basic processes like visual perception and spatial cognition (Henrich, Heine, & Norenzayan, 2010). The cultural intelligence hypothesis captures the argument of early theorists, and forwards that adult humans have a range of cognitive abilities not possessed by other animals, which are specially developed for their ways of life, and learned by an elaborated social learning process in development (Herrmann et al., 2007; van Schaik & Burkart, 2011). In addition to cognitive differences measurable in humans and animals, paleoanthropological evidence supports the extension of the childhood lifestage within humans compared to other apes (who have a comparatively quick juvenile period), consistent with the emergence of increasingly elaborate culture (Nielsen, 2012).

For these reasons, to understand cognition it is necessary to understand individuals as situated in their cultural context, and to understand the dynamics

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of culture: how it evolved, how information is transmitted between individuals and how modifications accumulate on cultural products. Acquiring such understanding poses methodological challenges as it requires examining both individual psychology and dynamics that arise from interactions between people. This chapter first reviews the current thinking on how to analyse both individual abilities for social learning and culture. It then reviews the diffusion experimental method, which allows an investigation of the relation between the individual and their culture, as well as the dynamics of cultural transmission. Finally, the chapter highlights a further area of complexity in cultural evolution research: the dynamic nature of social structure and culture. New approaches, and specifically new analytic methods, have recently been used to take into account the way in which culture flows through social relations. We propose that a bi-directional relationship between social learning and social relations also needs to be taken into account; as not only does social learning occur through social relations, but social learning in-and-of-itself creates and strengthens social bonds.

Analysing Culture: Social Learning Strategies

Laland (2004) laid out the social learning strategies framework, which seeks to understand how biases in what is learned might affect which cultural knowledge or products get propagated, and the adaptive basis of social information use (also see Rendell et al., 2011). This framework has become widely used in analysing social learning and culture. Laland (2004) draws on evidence showing that, while using social information can confer adaptive benefit over information the individual learns themselves, unconstrained use of social information (in the form of copying the behaviours of conspecifics) is not adaptive (Giraldeau, Valone, & Templeton, 2002). For instance, evolutionary modelling shows that unconstrained social information use can lead to the use of maladaptive out-of-date information within a population (Rogers, 1988). This begs the question: How did the widespread use of social information observed across species evolve? And specifically: How did the intensive use of social information seen within human culture evolve?

Social learning strategy research seeks to answer these questions by examining the factors which might contribute to the reliability of social information (often via modelling), and testing whether animals and humans adhere to these predictions. For instance, seeing a majority of individuals performing a behaviour may strengthen the reliability of that information – thus a 'copy-the-majority' bias in using social information may be adaptive (Pike & Laland, 2010). Prestigious individuals or dominant animals may be higher in the social hierarchy because of their more adept ability to use information – and thus copying the behaviour of these individuals may be adaptive (Chudek, Heller,

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Birch, & Henrich, 2012; Flynn & Smith, 2012; Henrich & Gil-White, 2001). It is posited that humans and animals may have adapted learning rules that confine social information use to instances where it is reliable. These learning rules help to explain how social information use arises and shed light on the differential transmission of information based on the architecture of our preferences in learning social information.

The social learning strategy framework aids in using these evolutionary constraints and dividing questions about social learning and cultural transmission into the logical components of 'what, whom, when and how?' *What* concerns the kind of information that is transmitted; *whom* examines the kinds of model that are preferred; *when* examines situations that promote social information use; and *how* addresses the mechanism of social learning, whether that be through imitation, language or other means.

Social learning strategies research has so far focused on several key avenues as best candidates for areas where social information may be useful (Rendell et al., 2011). The first is aspects of the source of social information or the model: such as their relation to the learner (Henrich & Henrich, 2010), their attributes such as age and/or knowledge state (Wood, Kendal, & Flynn, 2012), and their previous success (Mesoudi, 2008; for a review of whom-based biases, see Wood, Kendal, & Flynn, 2013). Second, frequency-based copying has also been investigated, including copying frequently observed behaviours (Toelch, Bruce, Meeus, & Reader, 2010) versus copying rare behaviours (Griskevicicus, Goldstein, Mortensen, Cialdini, & Kenrick, 2006). Attributes of the learner, such as their age, knowledge (van Bergen, Coolen, & Laland, 2004) or reproductive state (Webster & Laland, 2011), have also been shown to affect social learning. Certain informational content is especially likely to be transmitted – known as 'content biases' – such as information that causes a disgust reaction (Heath, Bell, & Sternberg, 2001) or draws on social rather than non-social information (Stubbersfield, Tehrani, & Flynn, 2014). The social learning strategy framework has been a productive way to organise research in cultural evolution. We now review the diffusion experiment method, which offers a particularly useful tool to investigate the link between individual cognition and culture, and measure the use, or not, of specific social learning strategies under certain contexts.

The Diffusion Experiment Methodology

Culture is a group-level phenomenon; it arises out of the action of many individuals, sharing and improving on information over time. Such dynamics produce methodological challenges for those interested in the psychological mechanisms and interpersonal processes which allow for culture. In many areas of psychology, participants can be tested in a laboratory setting individually or

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be surveyed *en masse*. However, to properly understand the relation between psychology and culture, it is necessary not only to understand the psychology of the individual, but also the interaction between individuals and group-level dynamics. There has been a plethora of studies examining social learning and information use in both humans and animals (Hoppitt & Laland, 2013; Zentall, 2006, 2012), but often this research has followed a 'dyadic' design. While this work has been essential in elucidating the capacities of different species and age groups of children in their ability and proclivities to use social information, it is limited in generalising about the dynamics of multiple individuals and how group-level behaviour gives rise to cumulative culture.

Previously, naturalistic observations either of populations of human cultural groups (DeWalt, Dewalt, & Wayland, 1998) or animals (Kawai, 1965) have been the most prominent method used to study cultural change. In naturalistic observation, a researcher attempts to describe how certain cultural products arise, how they are used, modified and spread through a natural population (e.g. a village or a troop of monkeys). However, naturalistic observation, being non-experimental, has limitations in identifying causality and discounting alternative explanations. Diffusion studies move the examination of the interactions of individuals and how they give rise to cultural processes into the laboratory, allowing greater experimental control and the possibility of manipulation (Flynn & Whiten, 2010; Mesoudi & Whiten, 2008; Whiten & Mesoudi, 2008). It should be noted that diffusion research, and the use of designs involving small groups, has its roots in social psychology and particularly research on group dynamics (Forsyth, 2006; Tajfel & Turner, 1986), which faces the same problem of examining the dynamic interaction of the individual and the group. In the case of social psychological research, the general concern is how groups influence the attitudes and behaviour of their members; whereas for research investigating 'cultural evolution', there is greater interest in how information spreads and changes. Although there are many iterations of experimental design and procedure which can be used in diffusion research, there are two overarching approaches: open diffusion and diffusion chains.

Open-diffusion designs (sometimes called the 'closed-group method'; Mesoudi & Whiten, 2008) are similar to the observation of cultural transmission in natural populations; however, the context in which the transmission occurs is controlled. Initially, a behaviour is seeded in a population, either by a pre-trained individual or by the experimenter creating the opportunity for the behaviour to arise and waiting until an individual discovers it. All individuals in the experiment have the opportunity to observe the interactions between other individuals and the task, and in principle to demonstrate this knowledge by acting on the task themselves. For instance, Flynn and Whiten (2010), in an initial phase, allowed children to play individually with an apparatus for which two discrete methods could be used to retrieve a reward. In a second phase, all

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children were allowed to interact in a single group. Variation in the use of the two methods by the children decreased over time within the group, compared to when they interacted with the apparatus individually. By videotaping the experimental session, Flynn and Whiten were able to examine the interactions between children, specifically how their history of observations of success and/or failure lead to this effect (see Figure 24.1).

In contrast to open-diffusion experiments, in diffusion chain experiments, the transmission process is controlled. The design works similarly to the 'Chinese Whispers' or 'Broken Telephone' games, such that an individual (A) is trained (or has the proclivity) to perform a certain type of behaviour, and then performs that behaviour in the presence of individual (B), and then individual (B) has an opportunity to perform the behaviour in the presence of another individual (C), and so on. Different chains are run to examine the fidelity of transmission along the chain of the seeded information according to their different characteristics – which can be manipulated, such as age or type of participant, content of behaviour or context. This design was first employed by Bartlett (1932), who seeded chains of adults with short stories which varied in their content. While details in the stories were often misconstrued or lost during transmission, by the end of the chain the essential idea or 'gist' of the story was retained. Bartlett further concluded that the distortions which occurred were based on the participants' prior knowledge.

Diffusion chain experiments can also involve a group of individuals in which certain individuals are removed and replaced, and in this way can attempt to replicate the process of cultural 'evolution' over several laboratory 'generations'. Groups perform a task, or are seeded with a behaviour or information, and then members are systematically withdrawn and replaced with new members. Jacobs and Campbell (1961) used this approach to extend Sherif's (1935) classic research on the 'autokinetic effect'. He had groups of three individuals judge the distance a dot of light moved in a darkened room, finding that the groups converged and agreed upon a common distance, thus demonstrating the effect of conformity. Jacobs and Campbell (1961) took the extra step of replacing one group member with a naïve participant, measuring the change in estimates over generations. It was found that norms persisted for several generations but eventually regressed back to the estimates of naïve individuals (also see von Zimmerman & Richardson, this volume, Chapter 12).

Using diffusion experiments allows control over many factors of the experimental design, but the most crucial for answering questions about cultural evolution is control over the manipulation of the group composition (which individuals are present) and the transmission or interaction over time (by controlling aspects of the generations within the experiment). How these aspects of design are varied provides different strengths and weaknesses, and the use of open diffusion and diffusion chains are suited to different questions regarding cultural evolution.

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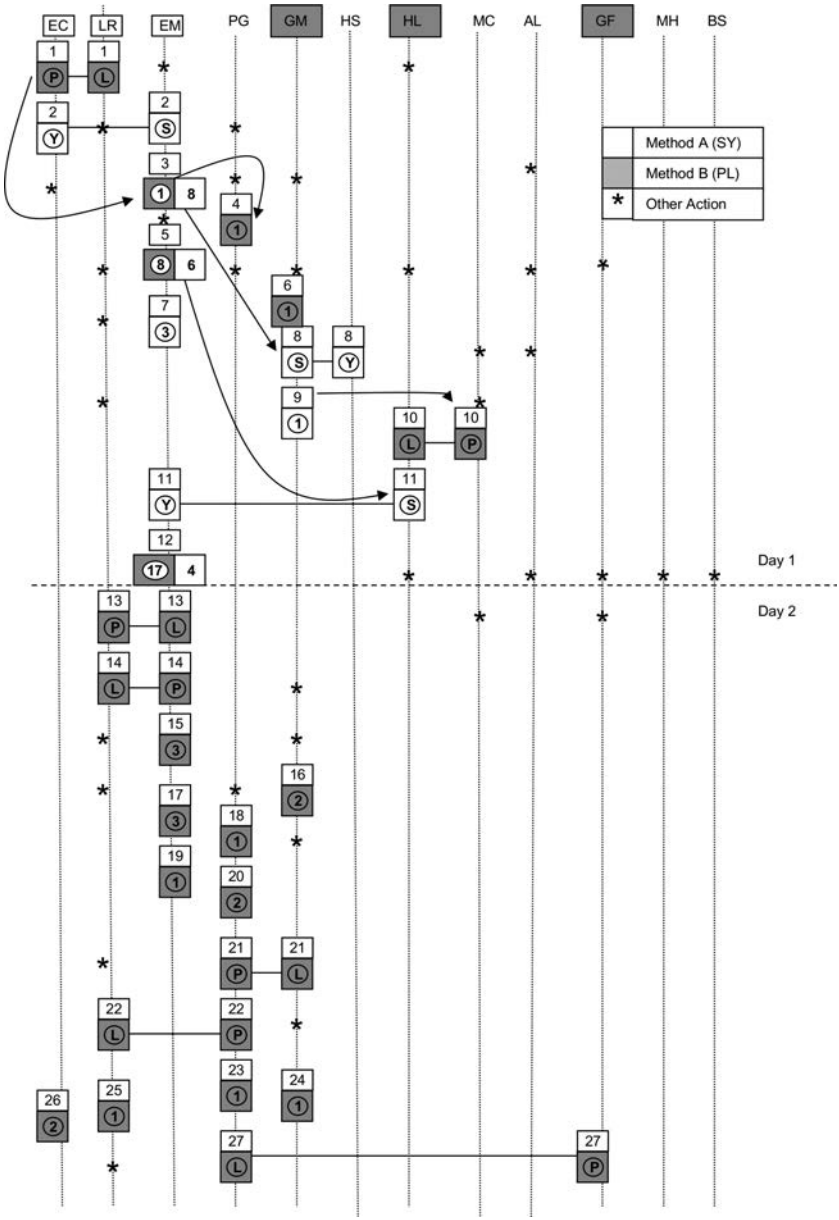

Figure 24.1 Open diffusion over two days.

Notes: The ordered columns are children's successes at extracting a reward from a puzzle box during the open diffusion. The initials in rectangles across the top are those children who were successful in an initial asocial phase;

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Figure 24.1 (*cont.*)

white rectangles represent use of method 'A', and dark rectangles represent use of method 'B'. No rectangle means that the child was not successful in the asocial phase. Rows represent successful bouts, sometimes using different methods to complete a series of successful extractions within the same bout. Arrows represent witnessing and adopting a new method, either for a child who had never been successful, or a change in method use in a previously successful child. Some attempts were joint attempts, represented by joining horizontal lines, with two children completing different parts of the extraction (P, turn pink knob; Y, turn yellow knob; L, push long lever; S, push short lever). An asterisk represents a child's interaction with the SB during another child's successful bout; these interactions included wiggling a door, attempting to turn an already turned knob, push an already pushed lever and/or switching on lights.

Source: Taken from Flynn and Whiten (2010).

In open diffusion, group composition is controlled by the experimenter, primarily by influencing the number and knowledge state of the individuals, while the interactions between these individuals unfold naturally over time. Comparisons across several such groups also allow specific research questions to be addressed. Open diffusion has been particularly useful in comparing the spread of social information to the effects of individual (asocial) learning. For instance, Flynn and Whiten's (2010) study, mentioned above, allowed comparison between children's use of behaviour acquired individually in the first phase and the behaviour that was observed in a group setting. Thus it could be established that there was homogeneity of behaviour within the group due to social learning, as is characteristic of culture. Open-diffusion designs have, therefore, often been used in animal experiments where the focus is on whether behaviour can spread and be maintained by social learning at all (Whiten & Mesoudi, 2008). How the interactions between individuals lead to changes in the frequency of a behaviour in a group can be measured, but as they arise organically analysing this is often 'messy' and difficult; yet, open diffusion has an advantage in terms of generalisability because of the natural spread of information it allows (Flynn & Whiten, 2010, 2012; Whiten & Flynn, 2010).

In basic diffusion chain designs where individuals interact sequentially over generations, the group composition is held constant (usually at dyads of two individuals), and the research question is generally the differential spread of information through chains or the specific interaction (the social learning mechanism) between individuals that leads to this information transmission. By seeding different forms of information in separate chains and measuring the fidelity or speed of information flow along the chain, the linear transmission chain method is particularly useful in examining biases which might exist in the spread of information (i.e. relate to which types of information are preferred

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and therefore spread quickly and faithfully). For instance, Mesoudi, Whiten, and Dunbar (2006) seeded chains of individuals with either information about a third-party ('gossip'), an individual's actions or the physical environment. Information about third parties was retained with the greatest accuracy, supporting a bias for content about social relations, and shedding light on culture by evidencing a hypothesis that humans have specialised social intelligence, for monitoring social relations.

In diffusion chain experiments in which the composition of groups is changed over time by withdrawing and replacing members, the experimenter exerts control both over the group composition and the generations within the experiment. These designs allow for both the study of intragroup dynamics, and how these intragroup dynamics affect the formation of cultural traditions, as well as the successfulness of these traditions. In this way, replacement chain experiments offer a powerful means by which to test hypotheses about cumulative culture. In the method used by Jacobs and Campbell (1961), the experimenter was able to observe how different intragroup processes, such as adherence to norms, contribute to the formation of knowledge within any generation, and how new members either influence or are influenced by these intragroup processes. Replacement experiments also allow the examination of the effect such processes have on the quality of the cultural products produced. For instance, Caldwell and Millen (2008) found that paper aeroplanes flew further and spaghetti towers were taller, over the course of multiple generations, due to the accumulation of information within groups. And Caldwell and Millen (2009) were able to test directly the association between different forms of available social information and learning, and the resulting effect these had on the transmission and accumulation of modifications on culture across generations. In Caldwell and Millen's (2009) study the amount of information available in separate chains within a paper aeroplane constructing task was varied; participants variously (1) had access to the products (paper aeroplanes) of the previous generation, (2) were able to view the previous generation creating their products, or (3) were taught by the previous generation – or combinations of these processes. All forms and combinations of the social information led to cumulative culture, challenging previous assumptions that only certain social learning mechanisms lead to cumulative culture (Tomasello et al., 1993). By using a diffusion design the link between social learning interactions and the generation of culture could be tested directly rather than only conjectured.

Diffusion experiments are increasingly being employed in studies of social learning and culture to offer empirical tests of hypotheses about the evolution of culture, how culture arises out of social learning and the factors which affect the differential spread of cultural information. According to two 2008 reviews, 34 diffusion studies were conducted with adults (Mesoudi & Whiten, 2008) and 33 with animals (Whiten & Mesoudi, 2008); a 2010 review identified 5 studies with children (Flynn & Whiten, 2010). This number has subsequently

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grown. By allowing control and close measurement of group composition and the transmission of information, diffusion experiments offer a flexible and powerful method linking dyadic research on social learning with research on the group processes underlying culture.

Dynamic Social Networks and Cultural Diffusion

While diffusion experiments offer a critical method for testing the relation between social learning and culture, social structure and social relations are important to culture in natural populations, and this creates a further level of dynamism. Social information flows between individuals, and therefore the relationship between individuals is key to understanding the transmission of information and culture. In fact, it has been argued that changes in social structure were important in allowing the development of cumulative culture; it is reasoned that greater population density leads to greater retention of cultural information, allowing it to persist across generations and accumulate modifications (Powell, Shennan, & Thomas, 2009; Sterelny, 2011). This insight, that social structure impacts cultural transmission, has opened up the recent use of network-based modelling, a promising method that allows the relationship between social structure and information flow to be investigated statistically (Cantor & Whitehead, 2013; Hoppitt & Laland, 2011). However, in examining the transmission of information between individuals along with the social structure, there is further complexity which must be accounted for: social learning, the exchange of information between individuals, itself changes the social structure. That is, there is a feedback loop between social structure and social learning. Copying others' behaviour serves both an informational function, of acquiring knowledge, and an interpersonal function, increasing affiliation and liking between the model and learner (van Baaren, Janssen, Chartrand, & Dijksterhuis, 2009; Chartrand & Bargh, 1999; Over & Carpenter, 2012; also see Oostenbroek & Over, this volume, Chapter 7). We suggest that this second 'affiliatory' function of social learning likely remodels the social network and can be implicated in the clustering of like individuals observed in social network research (Cantor & White, 2013; McPherson, Smith-Lovin, & Cook, 2001). To understand the evolution and dynamics of culture it is, therefore, necessary to investigate not only the effect of social structure, but the change in social structure and relations over time. We call attention to the developing method of social network analyses to be used in concert with diffusion experiments as a further way of investigating the dynamics of culture (a further method which might be fruitful, especially within the developmental literature, is the 'microgenetic' method, reviewed by Flynn & Siegler, 2007).

Social learning has a social role. In both developmental psychology (Over & Carpenter, 2012; Užgiris, 1981) and social psychology with adults (van Baaren et al., 2009; Chartrand & Bargh, 1999), the dual role of social learning as a method to gain accurate information and to increase affiliation has been a

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source of interest. Matching or conforming behaviour, which sends the message 'we are alike' (whether intended or unintended and unconscious), has been shown in a range of domains with both children and adults to increase affiliation (van Baaren et al., 2009; Chartrand & Bargh, 1999; Over & Carpenter, 2012; also see Lakens, Schubert, & Paladino, this volume, Chapter 13).

A demonstration of the social role of copying for children was shown by Nielsen and Blank (2011). Four- and five-year-old children observed two models using an apparatus, one who included an obviously causally redundant action when retrieving a reward from inside the apparatus, and one who used only causally necessary actions. The researchers then manipulated which one of the models left the room. During the children's opportunity to retrieve the reward, they were more likely to include the causally redundant action when the model who demonstrated it was present, as opposed to absent; suggesting that children were including that action for interpersonal reasons. Similarly, Over, Carpenter, Spears, and Gattis (2013) showed children two models, one of whom mimicked the children's choices in selecting which animal she liked best (depicted on photographs shown to the child), the other selecting a different choice. Children were more likely to endorse both the later preferences and factual claims of the model who mimicked them, rather than the model who did not.

For adults, the use of copying to promote 'alikeness' has been shown to be quite subtle (van Baaren et al., 2009). When individuals engage in interactions with a partner they copy the posture, emotions, mannerisms and speech patterns of that person, and this mimicry increases when the goal is to create affiliation with a partner (Lakin & Chartrand, 2003). This, indeed, has the effect of increasing liking (Chartrand & Bargh, 1999). Participants, naïve to the goal of the experiment, interacted with a partner who either mimicked or did not mimic their behaviour. Participants rated the partner who mimicked them as more likable, and the interaction as more smooth and harmonious than that with partners who did not mimic them.

The liking of individuals who are similar to you, and the copying of their behaviour holds for more abstract domains than just immediate behaviour. For instance, people are more attracted to confederates with similar birthdays and surnames as their own (Jones, Pelham, Carvalo, & Mirenberg, 2004). Group identification is an especially important and powerful level at which this relation between 'acting alike' and 'being liked' occurs; individuals are both pulled to conform to the behaviour and attitudes of their group, and showing these group behaviours increases liking among group members (Deutsh & Gerard, 1955; Tajfel & Turner, 1986).

'Homophily', the clustering together of individuals who are alike, has been well attested to in the social network literature (McPherson, Smith-Lovin, & Cook, 2001). Individuals with similar attributes, such as educational attainment, class background, preferences and beliefs, are more tightly connected,

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interacting with each other more often, than those to whom they are dissimilar. We propose that the dual role of copying, as both an information transfer mechanism and an affiliation-increasing mechanism, has a role in creating this clustering, and further highlights the necessity of examining (1) the social structure underlying social learning, and (2) the dynamic way in which this might change over time. Homophily is present in animals and predicts their interactions (e.g. Kutsukake, Suetsugu, & Hasegawa, 2006; Madden, et al., 2011; Sundaresan, et al., 2007), increasing interest in the question of the evolutionary basis of the phenomenon (Fu, Nowak, Christakis, & Flower, 2012). Below we review recent research that has employed network-based analyses as an innovative method to examine the relation between social learning and social structure.

Analysing the social network and the spread of social learning. Social network analyses are new methods being imported from the area of network theory to study how culture spreads through social learning within a social network (Cantor & Whitehead, 2013; Hoppitt & Laland, 2011). The analytic methods depict individuals as nodes in a network, with the connecting edges being the relationship between individuals. Edges can be weighted according to the closeness of those individuals, which, for example, may be measured by the proportion of time individuals spend together. This creates the network's topology. Social learning is then assumed to occur along edges, and thus learned traits to be transmitted quickly between closely linked individuals (see Figure 24.2 for an illustration). Networks, therefore, encode both the relation between individuals and the spread of transmission of information, and can in theory be used to model their interaction. In the area of cultural evolution, this method has so far been largely restricted to research on animals, yet these experiments demonstrate the promise of extending this method to other areas of 'culture'. For example, Mann and colleagues (2012) applied social network analysis methods to the spread of a tool-use behaviour within bottle-nosed dolphins. Within a population of dolphins in Shark Bay, Australia, there is differential use of sponges by individuals in foraging; while some dolphins hold the sponges on their beaks as they feed in rocky substrate, others do not. Using information about the time individuals spent together showed that sponge-using dolphins preferentially associate with other sponge users, supporting the hypothesis that the behaviour is related to social structure. Similarly, Allen, Weinrich, Hoppitt, and Rendell (2013) examined the foraging behaviour of humpback whales in the Gulf of Maine where two different foraging techniques exist: 'bubble-feeding' and a method that was an innovation in the population called 'lobtail feeding'. In bubble-feeding, whales blow bubbles below and around schools of prey fish forcing them together, allowing easier capture, but in lobtail feeding the tail is smacked on the surface of the water, and can be combined with bubble-feeding, to create the same effect. Using social network analyses, Allen and colleagues (2013) tested, and found support for, the hypothesis that association between individuals accounted for

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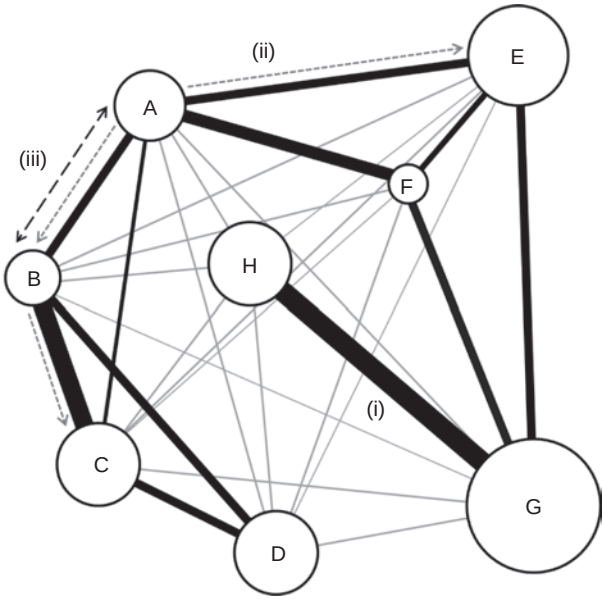

Figure 24.2 An illustration of a social network topology, as analysed by social network analyses.

Notes: Individuals appear as nodes (A, B, C…), which are connected by edges. Edges can be weighted to encode information about level of association (i). Culture/information spreads through edges (ii), and more quickly between more tightly associated individuals. We posit that the spread of social information/social learning itself has an interpersonal role, increasing affiliation, and therefore itself should rearrange network typology in a dynamic feedback process (iii).

the spread of the behaviour, finding that a model in which social transmission was responsible for the spread of lobtail feeding was the strongest supported.

Claidiere, Messer, Hoppitt, and Whiten (2013) combined open-diffusion and social network analyses in an experiment with squirrel monkeys. Two separate populations of monkeys were seeded with different techniques to retrieve a food reward from an apparatus and the amount of time individuals spent together at the site of the apparatus was measured. Group differences emerged, which were characterised by monkeys copying the seeded behaviour, and the behaviour was diffused through social learning. The eigenvector centrality, a measure of how connected an individual is to all other individuals, had a positive association with the speed at which an individual would acquire the foraging method seeded in their group, thus demonstrating how a position in a network influences social learning. It was further found that the initial group differences in

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traditions maintained by using the behaviour of the model eventually degraded over time, as individuals used non-seeded methods. This fits with a picture of cultural evolution wherein high fidelity transmission and close replication of individuals' actions is important in sustaining culture – abilities which are lacking in monkeys (Whiten, 2011).

These experiments begin to show how network analyses can help in linking together social relations and transfer of cultural knowledge, and testing theories about cultural evolution. They can also begin to tell us about mechanisms. For instance, Atton, Hoppitt, Webster, Galef, and Laland (2012) used network-based analyses to show that, while social network did predict three-spine sticklebacks' likelihood of finding a solution to a task, the social network did not predict the spread of the solution to the task. This suggests that the mechanism of transmission was, in fact, indirect, with individuals simply being present and more likely to discover the solution to the foraging task by being associated with knowledgeable fish and *not* direct observational learning.

The use of network-based analyses with animals has been developed largely because of the difficulty of evaluating hypotheses about social learning in natural animal populations – and has been employed generally in hope of discovering if social learning is present at all, and, if so, to what degree this leads to traditions or culture (Cantor & Whitehead, 2013; Hoppitt & Laland, 2011). The use of similar network-based statistics has not been readily employed within research into culture with humans, although there has been some modelling research investigating the relation between culture and social structure in systems displaying attributes similar to human populations (e.g. Centola, González-Avella, Eguíluz, & San Miguel, 2007). This modelling work suggests that, with humans, these methods could be used to examine subtler questions about the transmission of information, and the relationship between individuals, social structure and the adaptiveness and efficiency of the entire network (Cantor & Whitehead, 2013). Two studies, described below, give examples of the way in which network modelling can be used in the human case to elucidate answers to questions about culture.

First, Kunst and Kratzer (2007) examined the diffusion of an online product through groups of primary school-aged students in The Netherlands. They surveyed the relationships between children in these groups using questionnaires, and measured the uptake of the on-line product over time. Employing social network-based statistics, they were able to determine that children's centrality to their network (their strength of connection to all other individuals) was a strong determinant of their uptake of the product.

Second, Wisdom, Song, and Goldstone (2013) had groups of participants play a computer-based task where they were required to combine virtual pets into teams to compete in leagues. Within this game, participants received information about the various payoffs of combinations of virtual pets and had to

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attempt to maximise this payoff. Participants also had access to the solutions of other individuals in the group. This created a situation in which there was a complex problem space, where each individual was able to use social information, but also wished to increase her own payoff. Participants used social information in a range of ways, but specifically they used peers' solutions which were similar to their own. By imitating these solutions it was possible for further incrementally improved solutions to be developed, a signature of cumulative culture.

Culture is transferred via social learning through the social relations within a group's social network. Both humans and animals group together based on their likenesses and similarities. We have strong evidence within humans that social learning, including copying the behaviours of and adopting the preferences of others, has an interpersonal function of increasing liking, as well as the function of providing accurate information about the world. Together these points suggest that both social structure and its change over time are fundamental in understanding culture, and that there is a feedback-like relation in which social information transfer itself remodels the topology of the social network (Cantor & Whitehead, 2013). The use of social network analyses has already begun to prove successful in examining the relation between social structure and information flow within animals, and holds promise for similar investigations in humans. Especially promising are designs which couple social network analyses with the use of diffusion experiments, given the control they provide.

Conclusion

Cognition, especially human cognition, must be understood with reference to its situated nature within development, within a culture. The analyses of culture and social learning via social learning strategies have helped frame research into areas of logical investigation, and our knowledge of the situations in which social information is used and the mechanisms which allow this transmission is growing. However, much of this research has employed a dyadic design. To properly investigate culture it is necessary to examine the interactions between multiple individuals and how they give rise to the dynamics of culture. Diffusion experiments allow the direct testing of hypotheses about such dynamics by allowing the experimenter to manipulate the group composition and transmission of information over time. Further, the dynamic bi-directional relation between social network structure and the flow of cultural information must be explored. There is now substantial evidence that social learning provides functional information about one's environment *and* has an interpersonal function, ultimately changing social relationships. It is essential that we study this feedback relationship to establish how social learning

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restructures the social network. Social network analyses offer promise in investigating this process, being especially useful when combined with diffusion experiments.

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Glossary

Culture An amalgamation of socially-learned behaviour (traditions), which is often operationalised as behaviour that varies between groups but which is not attributed solely to differences in genetics or ecology.

Cumulative culture Culture exhibited by modern humans, in which learned information is modified or improved on, generation after generation, to allow ever increasing adaptive fit with the environment.

Dyadic design Experiments in which dyads (groups of two) interact. In the study of social learning, dyadic experiments often involve the interaction between a model and a learner.

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Homophily The tendency for individuals to affiliate with similar others.

Social information Information that is produced by conspecifics and is available. 'Social information use' refers to learning and using this information, and so social information use can be used synonymously with 'social learning'.

Social learning The process of learning from another individual; this can be through simple processes such as observation or more complex teaching.

Social learning mechanisms The specific process which allows the social learning to occur, such as 'imitation', where a learner watches and copies the actions of a model to produce the same outcome, or 'local enhancement', where simply being near another individual makes more likely the discovery of a behaviour.

Social learning strategy A rule regarding the particular circumstances under which social information is used that is evolutionarily adaptive, such as learning from reliable individuals.

Transmission The process of information being passed between individuals. In the context of the current chapter 'transmission chain' refers to information being passed from individual to individual, to individual, and so on.

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